

Digital Electronics : Design of a 4 bit Moebius counter

Abstract

This report explains what are counters and how they work and the various terms related to it. First we will take about simple counters and then talk about working of Moebius counter.

I. Counter

A digital counter is a set of flip flops whose states change in response to pulses applied at the input to the counter. In binary counter, the flip flops are interconnected such that their combined state at any time is the binary equivalent of the total number of pulses that have occurred up to that time. Thus a counter is used to count pulses. Each of the counts of the counter is called state of the counter. The number of states through which the counter passes before returning to the starting state is called modulus of the counter. In general, an n-bit counter will have 'n' flip flops and 2^n states and divides the input frequency by 2^n . Hence it is a divide by 2^n counter.

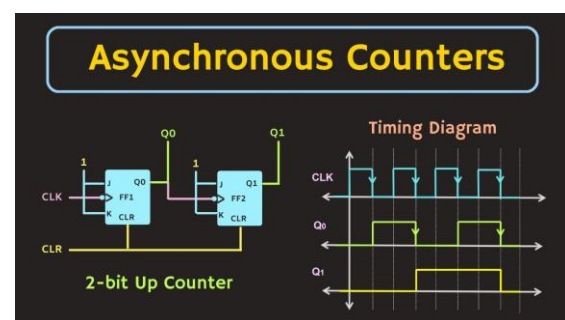
Each block prepares the signal so the microcontroller can analyze vibrations inside the rocket.

2. Clocks

There are two type of counter-

Asynchronous and synchronous.

Asynchronous means clock is not shared and synchronous clock means clock is shared. Each flip flops get a shared counter. A typical counter (riplle counter) is based on asynchronous clocks and count pulses in binary (0 , 1 , 2 , 3 and so on). N bit counter will have n flip flops.



Flip flop can be either rising or falling edge and to make ripple up/dow counter we use different combination of output (Q OR Q') and rising/falling edge.

3. Why a moebius was made and used practically

Disadvantages of binary counter-

1. Glitches

When a binary counter changes from 0111 to 1000 all 4 bits of the binary counter change at the time. But the thing is, the transistors in the binary counter do not switch on and off at the same moment. There are tiny delays. So for a short time the output of the binary counter goes through wrong states like 0110, 0100 0000 before it finally settles at 1000. Any logic circuit that is watching the output of the binary

counter during that time sees nonsense. This causes errors in the chips of the binary counter.

2. Ripple delay

In a binary counter the clock signal moves from one part to the next. FF0 triggers FF1 FF1 triggers FF2 and so on. Each part of the binary counter adds a delay. In a 16-bit binary counter that is 16 delays added together. At frequencies this becomes a big problem, with timing.

3. Decoding is complex

To detect a count value in a binary counter you need to use special gates that check multiple bits of the binary counter at the same time. These gates are called multi-input AND gates. They check multiple bits of the binary counter simultaneously.

Advantage of Moebius -

1. The thing about Gray code is that only one bit changes per clock. This gets rid of glitches completely. There is never a moment when two bits are switching at the time so you do not have to worry about wrong intermediate states. This was really important in digital hardware because timing margins were very tight.

flip-flops in a Gray code counter share the clock, which means there is no ripple delay at all. The whole counter updates in one clean clock edge no matter how many bits wide it is.

2. The wiring for a Gray code counter is extremely simple. You just chain the flip-flops in a ring. Invert one feedback wire. You do not need any logic gates all. This is great because it means you need components, which saves cost and you can make smaller chips. This was hugely valuable when transistors were expensive.
3. It is easy to decode each state in a Gray code counter. Every state in a Gray code counter can be decoded using a 2-input AND gate because each state has a unique pair of adjacent bits, in the Gray code counter. Example-

State	Q3Q2Q1Q0	Decode with
0	0000	$Q3'Q0'$
1	0001	$Q1'Q0$
2	0011	$Q2'Q1$
3	0111	$Q3'Q2$
4	1111	$Q3Q0$
5	1110	$Q1Q0'$
6	1100	$Q2Q1'$
7	1000	$Q3Q2'$

4, Working of Moebius counter-

The Möbius counter goes by several names. It is most commonly called the Johnson counter, named after engineer Robert Royce Johnson who patented the design in 1953. It is also called a twisted ring counter because it is structurally

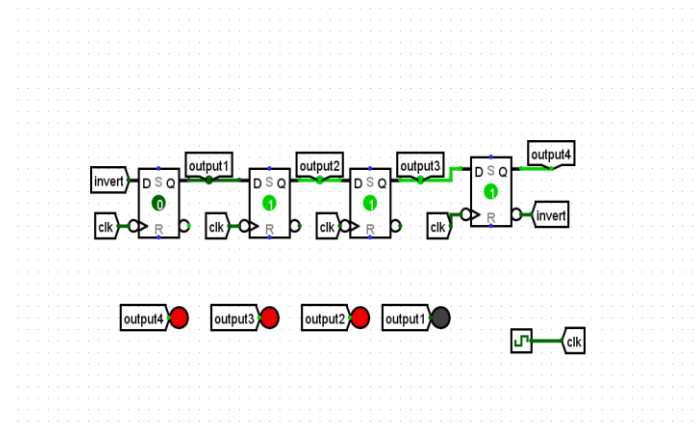
1. Gray code counters are synchronous by nature. All the

identical to a ring counter except for one inversion in the feedback path — that single twist is what gives it the Möbius name, inspired by the Möbius strip in mathematics which is a surface with only one side created by giving a strip of paper a half-twist before joining the ends.

Since we need gray code transitions , we give negative (inverted) feedback of last flip flop to first flip flop since we need to remove some states. This is an example of 3 bit. Mod means no. Of bits till first bit is repeated. Since this is 3 bit , MOD is $2n = 6$.

State / Number of clock pulses	State at moment t	State at moment $t+1$	Observations
Bits $\rightarrow$	Q_2, Q_1, Q_0	Q_2, Q_1, Q_0	
state 0 / "1"	0 0 0	0 0 1	
state 1 / "2"	0 0 1	0 1 1	
state 2 / "3"	0 1 0	x x x	Redundant state
state 3 / "4"	0 1 1	1 1 1	
state 4 / "5"	1 0 0	0 0 0	
state 5 / "6"	1 0 1	x x x	Redundant state
state 6 / "7"	1 1 0	1 0 0	
state 7 / "8"	1 1 1	1 1 0	

In the given report , J AND K flip flop is used but for my design i am using D flip flops to make 4 bit Moebius counter.



Here LEDS are used to symbolise bits, Left one is the 4th bit. D flip flops is used and a common clock. Flipping of last bit then changes the next flip flop value at the next FALLING edge and so on the bits change. This keeps happening due to supplying inverting bit of last flip flop output to first flip flop input. Each flip flop output is connected to LED to symbolize bits. This was made in logism evolution.

Clock	Q3	Q2	Q1	Q0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	1
3	0	1	1	1
4	1	1	1	1
5	1	1	1	0
6	1	1	0	0
7	1	0	0	0
8 (reset→0)	0	0	0	0

